

WINTER ARC

Digital 90-Day Self-Improvement Tracker

MASTER PROJECT SPECIFICATION

Build a polished, motivating, local-first web application that turns a user's Winter Arc into a measurable 90-day system. It should feel like a premium personal dashboard, not a generic to-do list.

Item	Specification
Primary user	One person managing their own 90-day Winter Arc
Core model	90 days + customizable habits + daily completion + analytics
Persistence	Browser localStorage; no account required for V1
Design	Dark cinematic / premium / focused / minimal
Build principle	Functional first, polished second; no unnecessary complexity

IMPORTANT FOR FREEBUFF

Treat this document as the source of truth for the first implementation. Do not invent extra product features, replace the core architecture unnecessarily, or turn the tracker into a social network.

1. Product Vision

Winter Arc Tracker is a personal 90-day discipline and self-improvement dashboard. The user defines what their Winter Arc means, chooses the habits they want to track, and checks progress every day. Daily actions become measurable streaks, completion percentages, perfect days, trends, and a visual 90-day map.

The product should answer these questions immediately:

- What do I need to do today? — today's habits and targets.
- How am I doing? — today's completion, current streak and quick stats.
- Am I consistent? — trend and habit-level analytics.
- Where am I in the Arc? — Day X / 90 and the full 90-day overview.
- Why am I doing this? — personal goal, why and rules.

The emotional direction is disciplined and calm. Avoid childish gamification, rainbow dashboards, fake achievements, social feeds, and distracting animation.

2. Core User Journey

Step	User action	Application response
1	Open first time	Show onboarding/setup.
2	Set Arc	Default 90 days; choose start date.
3	Define identity	Save goal, why and rules.
4	Add habits	Create customizable daily tracking items.
5	Open Dashboard	Show current day and today's habits.
6	Complete habits	Update progress, streaks and stats.
7	Review calendar	See completed / partial / missed days.
8	Review analytics	See consistency and habit performance.
9	Reflect	Store weekly reflection.
10	Edit Arc	Allow safe changes without destroying history.

First-launch rule: the entire V1 must work without login, backend, Firebase, or internet. A user can create and use the Arc locally.

3. Screens & Responsibilities

Dashboard / Home

- Hero: WINTER ARC, Day X / 90, dates and progress.
- Today's Tasks with completion controls.
- Today's Progress circular indicator.
- Quick stats: current streak, best streak, perfect days, consistency.
- 90-Day Overview grid.
- Responsive desktop cards and clean mobile stacking.

Today

- Focused daily view.
- Checkbox, numeric and duration inputs.
- Completion percentage and remaining habits.
- Optional daily note.
- Previous days viewable; future days protected from accidental completion.

Calendar / 90-Day Overview

- All 90 days chronologically.
- Completed, partial, missed, today and future states.
- Click a day for details.
- Do not rely on color alone.

Habits

- Add, edit, archive and reorder.
- Checkbox, numeric target and duration target.
- Archived habits preserve history.

Stats

- Overall completion, current/longest streak, perfect days.
- Habit rates and 7/30/90-day trends.
- All values derived from real data.

Reflection

- Weekly: what went well, what to improve, next focus.
- Store by week/date.

My Winter Arc

- Goal, why, rules, dates, status.
- Edit without deleting daily history.

4. Habit System — Detailed Logic

Every habit is a reusable definition. Daily completion is stored separately so editing a habit name does not rewrite history.

Type	Example	Completion rule
Checkbox	Read	Completed when checked.
Numeric	Drink water	Completed when actual $\geq$ target.
Duration	Study / work	Completed when minutes $\geq$ target.

Recommended habit object

```
{
  id: "habit_001",
  name: "Workout",
  icon: "dumbbell",
  type: "checkbox",
  target: 1,
  unit: "session",
  active: true,
  createdAt: "...",
  order: 1
}
```

Daily record example

```
{
  date: "2026-10-05",
  habits: {
    "habit_001": { value: 1, completed: true },
    "habit_002": { value: 6, completed: true }
  },
  note: "Good day",
  updatedAt: "..."
}
```

Do not hard-code targets such as 7 liters or 1 workout. Examples may be shown during setup, but the user chooses the actual target.

5. Local Storage Architecture

Use localStorage for V1. Centralize all storage operations in one service/module.

Key	Purpose
winterArc.settings	Theme, preferences and configuration.
winterArc.arc	Dates, goal, why, rules and status.
winterArc.habits	Habit definitions.
winterArc.dailyRecords	Daily completion records and notes.
winterArc.reflections	Weekly reflections.
winterArc.version	Schema version for migrations.

Root Arc object

```
{
  id: "arc_current",
  title: "Winter Arc",
  startDate: "2026-10-01",
  endDate: "2026-12-29",
  durationDays: 90,
  goal: "...",
  why: "...",
  rules: ["...", "..."],
  status: "active",
  createdAt: "...",
  updatedAt: "..."
}
```

Persistence requirements

- Persist meaningful changes immediately.
- Reloading/reopening the browser restores progress.
- Handle missing/corrupt storage safely.
- Export complete app state as JSON.
- Validate imports before replacing current state.
- Reset All Data requires strong confirmation.

6. Progress & Analytics Rules

Metric	Definition
Day number	$\text{floor}(\text{today} - \text{startDate}) + 1$, clamped to 1..90 while active.
Daily completion	$\text{completed eligible habits} / \text{eligible habits} \times 100$.
Perfect day	All active habits completed for that day.
Current streak	Consecutive recent qualifying days.
Best streak	Longest consecutive run of qualifying days.
Habit rate	$\text{Completed occurrences} / \text{eligible occurrences} \times 100$.
Arc progress	Elapsed Arc days / 90 for the journey visual.

Initial streak rule

A streak day is a day where all active habits scheduled for that day are completed. Future days never break a streak. Calculate the current streak backward from the latest qualifying day through consecutive qualifying dates.

Keep formulas in a dedicated analytics service so rules can change later without rewriting UI code.

7. UI / UX Design System

Area	Direction
Background	Near-black / deep navy.
Cards	Slightly lighter navy-black with subtle borders.
Primary accent	Icy blue / cyan.
Success	Reserved green for completed states.
Typography	Strong modern sans-serif hierarchy.
Corners	Moderately rounded and consistent.
Shadows	Subtle; avoid excessive glow.
Motion	Short, purposeful transitions only.
Icons	Consistent outline/duotone family.

Hierarchy

- Header first: Day X / 90 and progress.
- Today's work is primary.
- Analytics are secondary but visible.
- 90-day map gives long-term context.
- Reflection and Arc identity should not compete with today's tasks.
- Mobile: large touch targets, clear spacing, no horizontal overflow.

Avoid

- Overloaded gradients and excessive colors.
- Large decorative images consuming functional space.
- Social feeds, leaderboards and competition.
- Modal dialogs for every tiny action.
- Tiny controls on mobile.

8. Recommended Architecture

```
src/  
components/  
layout/ Sidebar, MobileNav, PageHeader  
dashboard/ ArcHero, TodayTasks, TodayProgress, QuickStats, NinetyDayGrid  
habits/ HabitList, HabitCard, HabitForm, HabitProgressInput  
calendar/ ArcCalendar, DayDetail  
stats/ StatCards, CompletionChart, HabitAnalytics  
reflection/ ReflectionForm, ReflectionHistory  
arc/ ArcProfile, RulesEditor  
settings/ DataManagement  
pages/ Dashboard, Today, Calendar, Habits, Stats, Reflection, MyArc, Settings  
services/ storage, analytics, date, exportImport  
hooks/ useArc, useHabits, useDailyRecords, useAnalytics
```

Architecture principle

UI components render state. Business logic belongs in services/hooks. Storage logic belongs in the storage service. Analytics belongs in pure, testable functions.

9. Onboarding & Setup

Screen	Fields / action
Welcome	Short explanation + Start My Arc.
Arc Setup	Start date and duration (90 default).
Identity	Goal and Why.
Rules	Add 3–7 personal rules; optional.
Habits	Create initial habits with editable examples.
Finish	Open Day 1 dashboard.

Editing rules

- Goal/why/rules edits never erase history.
- Start-date change requires confirmation.
- Duration change after progress requires confirmation.
- Deleting a habit means archive by default.
- Only Reset All Data removes the whole Arc.

10. Edge Cases & Reliability

- No data → onboarding.
- Missing storage → safe defaults.
- Invalid JSON → clear rejection.
- Duplicate IDs → normalize imported IDs.
- Use local calendar-date calculations.
- Month/year boundaries and leap years must work.
- Future dates cannot be accidentally completed.
- Habit edits preserve history by habit ID.
- Archived habits remain in history.
- No habits → helpful empty state.
- Long names truncate visually but preserve full value.
- No mobile horizontal overflow.
- Refresh after actions retains state.

Accessibility

- Accessible labels for interactive controls.
- Keyboard navigation and visible focus.
- Never communicate state by color alone.
- Charts include text summaries.
- Comfortable mobile touch targets.

11. Implementation Requirements for Freebuff

Build in phases. Do not generate every feature at once and leave half-working screens.

Phase	Deliverable	Done when...
1	Project shell + theme	Responsive app runs.
2	Arc setup	90-day Arc persists after refresh.
3	Habit CRUD	Add/edit/archive/reorder works.
4	Daily tracking	Completion persists and updates immediately.
5	Dashboard	Primary cards use real data.
6	Calendar	Past/current/future states correct.
7	Analytics	Stats use real records.
8	Reflection	Weekly entries persist.
9	Import/export	Backup and restore work.
10	QA/polish	No console errors or broken states.

Technology guidance

- If the repository already uses React + Vite, continue with it.
- Keep dependencies minimal.
- No backend for V1.
- No Firebase just for storage.
- No authentication unless explicitly requested.
- No social or AI features in V1.

12. Definition of Done

- Create a new Winter Arc.
- Add at least five habits.
- Complete checkbox, numeric and duration habits.
- Reload and retain data.
- Navigate past/current/future days.
- See the 90-day overview update.
- Verify current and best streaks.
- Verify a perfect day.
- Archive a habit while retaining history.
- Edit goal/rules without losing records.
- Create/retrieve weekly reflections.
- Export all data.
- Clear and import the export to recover the same state.
- Test desktop and mobile viewport.
- Run production build successfully.
- No uncaught console errors.
- No fake demo progress.
- No placeholder buttons.

Demo data rule

Sample data can exist only as an optional demo mode. The normal dashboard must always be driven by stored user data. Never hard-code statistics such as 78% completion.

13. MASTER INSTRUCTION TO THE AI CODER

Copy the following into Freebuff/OpenCode after placing this PDF in the project folder.

You are implementing a production-quality personal Winter Arc Tracker.

SOURCE OF TRUTH:

Use the attached Winter Arc Tracker Complete Project Spec as the authoritative requirements document. Do not invent a different product.

GOAL:

Build a polished, responsive, dark cinematic 90-day personal habit tracker. It must work locally in the browser and persist all user data using localStorage.

CORE FEATURES:

1. First-run Winter Arc setup.
2. 90-day Arc with start/end dates.
3. Goal, why statement and personal rules.
4. Custom habits.
5. Checkbox, numeric-target and duration-target habits.
6. Daily completion tracking.
7. Dashboard with Day X/90.
8. Today's tasks and progress.
9. Current streak, best streak and perfect days.
10. Full 90-day overview/calendar.
11. Habit analytics and completion trends.
12. Weekly reflections.
13. My Winter Arc profile.
14. Import/export JSON backup.
15. Safe reset and data validation.

ENGINEERING:

- Continue the existing stack if the repository already specifies one.
- No backend for V1.
- Centralize storage operations.
- Keep analytics deterministic and testable.
- Never hard-code progress values.
- Never destroy history when editing a habit.
- Archive habits instead of deleting historical records.
- Persist meaningful actions immediately.
- Handle missing/corrupt storage safely.
- Make desktop and mobile responsive.
- No fake buttons or placeholder screens.

DESIGN:

Premium dark/navy interface, icy-blue primary accent, restrained green success state, subtle borders, strong typography, moderate rounded cards, minimal animation. Serious and focused, not childish.

IMPLEMENTATION ORDER:

Foundation -> Arc setup -> habits -> daily tracking -> dashboard -> calendar -> analytics -> reflections -> data management -> QA/polish.

BEFORE FINISHING:

Test fresh install, refresh persistence, mobile layout, habit CRUD, daily calculations, streak calculations, import/export, and production build. Fix all console errors and broken interactions.

DO NOT expand V1 into social features, accounts, cloud sync, leaderboards, chat, AI coaching, or unnecessary gamification.

If a requirement is ambiguous, choose the simplest implementation that preserves the core user experience and data integrity. Do not silently remove a requirement.

Final V1 goal: a person should open the app, understand what to do today within five seconds, complete the work, see honest progress, and trust the data.

14. Future Expansion — NOT V1

- Cloud sync / accounts.
- Multiple Winter Arcs.
- Advanced scheduling.
- Reminders / notifications.
- AI coaching.
- Shareable progress cards.
- Public/social challenges.
- Cross-device synchronization.
- Native mobile wrapper.

V1 PRIORITY

A single person should be able to open the app, understand what to do today, complete the work, see honest progress, and trust the data tomorrow.

END OF SPECIFICATION

Winter Arc Tracker • Master Build Document